

Calculus II Lesson 03

1) Find $\int (\cos^4 x) \cdot \sin x \cdot dx$

Hint: Let $u = \cos x$

a) $\frac{du}{dx} = \underline{\quad ? \quad}$ and $du = \underline{\underline{\quad ? \quad}} \cdot dx$

b) $\int (\cos^4 x) \cdot \sin x \cdot dx = \int \underline{\quad ? \quad} \cdot du$ (write the integral $\int (\cos^4 x) \cdot \sin x \cdot dx$ in terms of u)

c) $\int (\cos^4 x) \cdot \sin x \cdot dx = \underline{\quad ? \quad}$

2) Find $\int (\sin^5 2x) \cdot \cos 2x \cdot dx$

Hint: Let $u = \sin 2x$

a) $\frac{du}{dx} = \underline{\hspace{1cm} ? \hspace{1cm}}$ and $du = \underline{\hspace{1cm} ? \hspace{1cm}} \cdot dx$

b) $\int (\sin^5 2x) \cdot \cos 2x \cdot dx = \int \underline{\hspace{1cm} ? \hspace{1cm}} \cdot du$ (write the integral $\int (\sin^5 2x) \cdot \cos 2x \cdot dx$ in terms of u)

c) $\int (\sin^5 2x) \cdot \cos 2x \cdot dx = \underline{\hspace{1cm} ? \hspace{1cm}}$

3) Find $\int (\sin^5 x) \cdot \cos^2 x dx$

Hint: $\int (\sin^5 x) \cdot \cos^2 x dx = \int (\sin^2 x)(\sin^2 x) \cdot (\cos^2 x)(\sin x) dx$

Hint: $\sin^2 x = 1 - \cos^2 x$; let $u = \cos x$

a) $\frac{du}{dx} = \underline{\quad ? \quad}$ and $du = \underline{\quad ? \quad} \cdot dx$

b) $\int (\sin^5 x) \cdot \cos^2 x dx = \int \underline{\quad ? \quad} \cdot du$ (write the integral $\int (\sin^5 x) \cdot \cos^2 x dx$ in terms of u)

c) $\int (\sin^5 x) \cdot \cos^2 x dx = \underline{\quad ? \quad}$

4) Find $\int (\sin^3 4x) \cdot \sqrt{\cos 4x} dx$

Hint: $\int (\sin^3 4x) \cdot \sqrt{\cos 4x} dx = \int (\sin^2 4x) \cdot (\cos 4x)^{1/2} (\sin 4x) dx$

Hint: $\sin^2 4x = 1 - \cos^2 4x$; let $u = \cos 4x$

a) $\frac{du}{dx} = \underline{\hspace{1cm} ? \hspace{1cm}}$ and $du = \underline{\hspace{1cm} ? \hspace{1cm}} \cdot dx$

b) $\int (\sin^3 4x) \cdot \sqrt{\cos 4x} dx = \int \underline{\hspace{1cm} ? \hspace{1cm}} \cdot du$

(write the integral $\int (\sin^3 4x) \cdot \sqrt{\cos 4x} dx$ in terms of u)

c) $\int (\sin^3 4x) \cdot \sqrt{\cos 4x} dx = \underline{\hspace{1cm} ? \hspace{1cm}}$

5) Find $\int (\cos^3 7x) dx$

Hint: $\cos^3 7x = \cos^2 7x \cdot \cos 7x = \cos^2 7x (1 - \sin^2 7x) \cdot \cos 7x$

Hint: Let $u = \sin 7x$

Note: For integrals where the power of the cosine is odd:

1) Save a cosine factor

2) Convert the rest to sines

3) Use formula $D_x (\sin x) = \cos x$

a) $\frac{du}{dx} = \underline{\quad ? \quad}$ and $du = \underline{\quad ? \quad} \cdot dx$

b) $\int (\cos^3 7x) dx = \int \underline{\quad ? \quad} \cdot du$ (write the integral $\int (\cos^3 7x) dx$ in terms of u)

c) $\int (\cos^3 7x) dx = \underline{\quad ? \quad}$

6) Find $\int (\tan^3 5t)(\sec^3 5t)dt$

Hint: $\int (\tan^3 5t)(\sec^3 5t)dt = \int (\tan^2 5t)(\sec^2 5t)(\tan 5t)(\sec 5t)dt$

Hint: $\tan^2 5t = \sec^2 5t - 1$;

$\therefore \int (\tan^2 5t)(\sec^2 5t)(\tan 5t)(\sec 5t)dt = \int (\sec^2 5t - 1)(\sec^2 5t)(\tan 5t)(\sec 5t)dt$

let $u = \sec 5t$

Note: For integrals where the power of the tangent is odd:
 1) Save a secant-tangent factor
 2) Convert the rest to secants
 3) Use formula $D_x (\sec x) = \sec x \cdot \tan x$

a) $\frac{du}{dx} = \underline{\hspace{1cm} ? \hspace{1cm}}$ and $du = \underline{\hspace{1cm} ? \hspace{1cm}} \cdot dx$

b) $\int (\tan^3 5t)(\sec^3 5t)dt = \int \underline{\hspace{1cm} ? \hspace{1cm}} \cdot du$ (write the integral $\int (\tan^3 5t)(\sec^3 5t)dt$ in terms of u)

c) $\int (\tan^3 5t)(\sec^3 5t)dt = \underline{\hspace{1cm} ? \hspace{1cm}}$

7) Find $\int (\sec^6 2x)(\tan 2x)dx$

Hint: $\sec^6 2x \cdot \tan 2x = \sec^2 2x \cdot \sec^2 2x \cdot \sec^2 2x \cdot \tan 2x$

$$= (1 + \tan^2 2x) \cdot (1 + \tan^2 2x) \cdot \tan 2x \cdot \sec^2 2x$$

Hint: Let $u = \tan 2x$

Note: For integrals where the power of the secant is even:

1) Save a secant-squared factor

2) Convert the rest to tangents

3) Use formula $D_x (\tan x) = \sec^2 x$

a) $\frac{du}{dx} = \underline{\quad ? \quad}$ and $du = \underline{\quad ? \quad} \cdot dx$

b) $\int (\sec^6 2x)(\tan 2x)dx = \int \underline{\quad ? \quad} \cdot du$

(write the integral $\int (\sec^6 2x)(\tan 2x)dx$ in terms of u)

c) $\int (\sec^6 2x)(\tan 2x)dx = \underline{\quad ? \quad}$

